

Sales & Profitability Analysis

Overview

This project looks at sales and profitability throughout the year, focusing on **monthly trends, regions, product categories, and discounts**.

The business generated around **\$2.43M in sales and \$308K in profit**, with an overall profit margin of approximately **12.7%**.

The goal is to understand where the business is performing well, where profitability is weaker, and whether discounts might be playing a role.

Monthly Performance

Sales increased noticeably toward the end of the year. **November had the highest sales and order volume**, with \$353K in sales and 755 orders.

However, November did not generate the most profit. **December came first with \$44K in profit**, despite having slightly fewer orders.

February is another interesting case. With only 162 orders, it had the **highest profit margin of the year at 17%**.

Overall, the numbers show that **more orders do not necessarily mean higher profitability**.

Regional Performance

West was the strongest region overall, with around \$740K in sales and a 15% profit margin.

Central stands out for a different reason. It generated more sales than South, but its profit margin was only **8%**, compared with 12% in South.

This makes Central one of the main areas worth looking into, especially when we consider its discount levels.

Category Performance

The difference between categories is quite significant.

Technology and Office Supplies both have a 17% profit margin, while Furniture is much lower at only **3%**.

Furniture is therefore the main category to look into, particularly when it comes to pricing and discounts.

Discounts & Profitability

The discount data shows some interesting patterns.

Furniture has the highest average discount at 17% and the lowest profit margin at 3%.

We see a similar pattern across regions. **Central has the highest discount at 16% and the lowest margin at 8%**, while West has the lowest discount at 13% and the highest margin at 15%.

The monthly results are less consistent, so discounts are clearly not the only factor affecting profitability. Still, the relationship is strong enough to make discounting worth a closer look.

Key Takeaways

- Sales and order volume increased significantly toward the end of the year.
- More orders did not always translate into higher profitability.
- **West** is the strongest region, while **Central** has the lowest margin.
- **Furniture** is the main profitability concern, with a 3% margin.
- Higher discounts tend to coincide with lower margins in some of the weaker-performing segments.

Recommendations

- **Take a closer look at Furniture discounts**, which are the highest at 17%, while the category has only a 3% profit margin.
- **Review discounting in Central**, where a 16% discount comes with the lowest regional margin at 8%.
- **Use West as a benchmark**, since it combines the lowest regional discount (13%) with the highest margin (15%).
- **Test different discount ranges** to see how much additional sales volume is needed to make a higher discount worthwhile.

Conclusion

The business had strong sales growth throughout the year, especially toward the end, but higher sales did not always mean higher profitability.

Furniture and Central stand out as the main areas to look into, as both have relatively high discounts and low margins.